The Personal Income Tax

EC313 - Public Economics: Taxation

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Goals of This Section

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In this section we will:

- Outline the history of personal income taxes in Canada
- Walk through how tax liability is calculated
- Compare Haig-Simons income with the income Canada actually taxes
- Explain the difference between deductions and credits
- Describe Canada's tax rates, statutory and effective
- Discuss whether the individual or the family should be the tax unit
- Explain bracket creep and indexation for inflation
- Contrast territorial and global taxation of foreign income

A Brief History of Personal Income Taxation

Origins

- Income tax first introduced federally in Canada in 1917 to help finance WWI
  - Initially a temporary measure, but became permanent in 1949
- Provinces began introducing income taxes earlier, in the late 1800s (BC, PEI)
  - Most others followed after the Great Depression
  - Newfoundland was last to introduce income tax in 1949
- Currently, both federal and provincial governments levy personal income taxes
  - Authority comes from Constitution Act

Personal Income Tax Revenues

TABLE 17.1 Personal Income Tax Revenues in Canada, 1933–2019

Year	Number of Returns (thousands)	PIT Revenues (millions $)	Per Capita 2014 $*	PERSONAL INCOME TAXES AS % OF: Total Revenues	GDP
1933	52	38	60	5.1	1.1
1946	3,162	671	668	17.9	5.6
1955	4,923	1,318	682	19.2	4.6
1965	7,163	3,563	1,253	20.8	6.2
1975	12,002	18,538	3,300	28.4	10.7
1985	15,864	53,262	3,815	30.5	11.0
2000	22,237	143,951	5,833	30.7	13.4
2012†	25,453	164,692	4,970	23.7	10.0
2019	27,815	227,404	5,463	24.2	13.1

- All governments, 2022: about 36% of *tax* revenue, not all revenue as in the table (OECD average 24%), and about 12% of GDP
- Federal government, 2024–25: $234.3B, 45.9% of federal revenue and 7.6% of GDP

Administration and Filing

- The Canada Revenue Agency (CRA) collects income tax for Ottawa and every province except Quebec
  - Provinces accept the federal definition of income; Quebec residents file two returns
- Until 2000, provinces levied “tax on tax”: a percentage of federal tax owed
- By 2001 all had moved to “tax on income”: each applies its own rates and credits to federal taxable income
- The system relies on self-assessment: if you owe tax, you file by April 30
  - The self-employed file by June 15, but any balance is still due April 30
- Many people who owe nothing file anyway to collect the CCB, the CWB and the Canada Groceries and Essentials Benefit (formerly the GST/HST credit)
  - Even so, about 12% of working-age adults did not file (Robson and Schwartz, 2020)

iClicker Question

Under the old “tax on tax” system, a province’s income tax was 50% of the federal tax each person owed. Ottawa cuts everyone’s federal tax by one-tenth: someone who owed $10,000 now owes $9,000. What happens to that province’s income tax revenue?

A) Nothing, because the provincial rate is still 50%
B) It falls by one-tenth, without any decision by the province
C) It rises, because people have more after-tax income to spend
D) It falls only if the province decides to cut its own rates
E) The provincial rate falls from 50% to 45%, matching the federal cut

Computation of Personal Income Tax Liability

Calculating Taxes

- Six steps: total income, net income, taxable income, federal tax, minus credits, then provincial tax
- Not all income is taxed
  - Gifts and inheritances, lottery and gambling winnings, strike pay, legal settlements, First Nations income on reserve, gains on a principal residence, most refundable credits
- Deductions come off total income before the rates apply, so taxable income is smaller than total income
- Tax is withheld at source through the year
  - At filing you pay the difference or get a refund
- The main return is the T1, plus schedules: see the CRA tax packages

Steps 1–3: From Total Income to Taxable Income

STEP 1: Calculate Total Income
ADD: Income from taxable sources:

- Wages, commissions, tips, and other employment income
- Pension income including income from Old Age Security, Canada/Quebec Pension Plan, RRSPs, and other pension income
- Employment Insurance income
- Interest and other investment income
- Dividends (grossed-up)
- Rental income
- Income from partnerships
- Taxable capital gains
- Taxable support payments received
- Royalties
- Pension and employment insurance benefits
- Net self-employment income
- Provincial workers' compensation benefits and income assistance
- Net federal supplements including the Guaranteed Income Supplement

= Total Income (line 15000 on the tax return)

STEP 2: Calculate Net Income
SUBTRACT from total income certain deductions:

- Pension contributions including RPP, RRSP, CPP, and QPP
- Pension income transferred to spouse (pension income splitting)
- Union dues and certain employment expenses
- Childcare expenses
- Disability supports
- Allowable business losses
- Moving expenses
- Deductible support payments
- Interest expenses
- Other special deductions
- Any repayments to Old Age Security, Employment Insurance, and any economic recovery benefits (e.g., CERB)

= Net Income (line 23600 on the tax return)

STEP 3: Calculate Taxable Income
SUBTRACT from net income certain deductions:

- Canadian Forces personnel and policy deduction
- Employee home relocation deduction
- Security options deduction
- Provincial workers' compensation and income assistance benefits
- Net federal supplements
- Various losses
- Northern residents deduction
- Additional other special deductions

= Taxable Income (line 26000 on the tax return)

Steps 4–6: From Taxable Income to Tax Payable

STEP 4: Calculate Federal Tax Owing
APPLY: Tax rate schedule to Taxable income
= Income tax before tax credits

STEP 5: Calculate Nonrefundable Tax Credits
SUBTRACT from tax owing:
Nonrefundable tax credits for

- Taxpayer and dependents
- Medical expenses and disabilities
- Age and pension income
- Contributions to QPP and CPP
- Premiums paid for EI
- Eligible tuition and education expenses
- Gifts to charities and to the Crown
- Any other tax credits

= Federal tax payable

STEP 6: Calculate Provincial Tax Payable
REPEAT steps 4 and 5 for provincial taxes.

The Haig-Simons Definition of Income

Haig-Simons Income

Haig-Simons (H-S) Definition of Income

The money value of the net increase in a person's power to consume during a period: consumption plus the change in net worth

- Includes all potential consumption, spent or saved, in any form
  - Employer pension contributions and insurance
  - Transfer payments: CPP/QPP benefits, EI, workers' compensation, social assistance
  - Capital gains and losses: whether or not the asset is sold
  - Income in kind: employer perks, imputed rent on an owner-occupied home
  - Gifts and inheritances
- Anything that reduces the power to consume is subtracted, e.g., costs of earning income

Practical and Conceptual Problems

- Business expenses: only income net of costs adds to the power to consume, but the line is hard to draw
  - A home-office desk that is also good furniture; a business lunch that is also a nice meal (50% of meal costs are deductible)
- Capital gains: hard to measure until the asset is sold
  - Easy for traded shares; hard for art, wine, collectibles, crypto and NFTs
- Imputed rent: homes and other durables yield services nobody pays for
  - Every house is different; cars and boats yield services too
- Household production: valuable, but there's no market price
  - Cooking, child care, home repairs, yard work

Evaluating Haig-Simons

Fairness

- Horizontal equity needs every source of income in the base
  - Otherwise people with equal incomes pay different tax
- But equal income need not mean equal well-being
  - Same income from pleasant brain work or unpleasant physical work

Efficiency

- Taxing all income alike is neutral
  - Untaxed imputed rent pulls too much investment into housing
- But neutral is not least distorting
  - Inelastic sources like land rent can be taxed more with no excess burden (Chapter 16)

> **Bottom Line**
>
> H-S is a benchmark, not a rule: judge each departure on its merits

iClicker Question

Under the Haig-Simons definition, which of these is *not* part of Sam's income this year?

A) The rise in value of shares Sam bought last year and still owns
B) The rental value of the condo Sam owns and lives in
C) A $5,000 inheritance from Sam's grandmother
D) The $10,000 Sam takes out of savings to buy a car
E) Sam's employer's contribution to Sam's pension plan

How Canada’s Tax System Defines Income

Departures from Haig-Simons

- Canada's base is a hybrid of an income tax and a consumption tax
  - The Carter Commission (1966) and the 1971, 1981 and 1987 reforms moved it toward H-S
- Some income is left out, or taxed only later
  - Left out: imputed rent, employer-paid health and dental plans, gifts and inheritances
  - Taxed later: employer pension contributions, when the pension is paid
- Capital gains get special treatment
  - Only half of a gain is taxed, and only when the asset is sold (realized)
  - Capital losses can only offset capital gains
- Exemptions remain
  - Gains on a principal residence are not taxed
  - Lifetime exemption on farm, fishing and small-business shares: $1,275,000 in 2026

Realization: Gains Taxed at Sale

- Buy an asset for $100; it grows 20% a year and is sold after 5 years
- Half of a capital gain is taxable, at a tax rate of 50%
- Realization (Canada): no tax until the asset is sold

Year	Start of year	Gain (20%)	Taxable gain (half)	Tax (50%)	End of year
1	100.00	20.00	0.00	0.00	120.00
2	120.00	24.00	0.00	0.00	144.00
3	144.00	28.80	0.00	0.00	172.80
4	172.80	34.56	0.00	0.00	207.36
5	207.36	41.47	0.00	0.00	248.83

- Sold in year 5: gain $148.83, taxable half $74.42, tax at 50% $37.21
- Left after tax: 248.83 − 37.21 = **$211.62**

Accrual: Gains Taxed Every Year

- Same asset, but accrual (Haig-Simons) taxes each year's gain that year, sold or not
- The tax comes out of the investment, so each year starts from a smaller value

Year	Start of year	Gain (20%)	Taxable gain (half)	Tax (50%)	End of year
1	100.00	20.00	10.00	5.00	115.00
2	115.00	23.00	11.50	5.75	132.25
3	132.25	26.45	13.23	6.62	152.08
4	152.08	30.42	15.21	7.61	174.89
5	174.89	34.98	17.49	8.75	201.12

- Left after tax: **$201.12**, which is **$10.50** less than under realization
- Less tax in total ($33.73 vs $37.21), but only because the gains are smaller ($134.85 vs $148.83)

Deferral, Lock-In and Deemed Realization

- Taxing gains only at sale defers the tax
  - The deferred tax is an interest-free loan from the government, so the asset grows at the pre-tax return
  - E.g., the $5 of tax not paid in year 1 earns $1 in year 2 ($0.75 after tax)
  - Hence the accountants' line that taxes deferred are taxes saved

> **Lock-In Effect**
>
> Because selling triggers the tax, owners hold on to assets they would otherwise sell, so capital doesn't flow to its best use

- At death, gains are deemed realized: taxed as if the assets were sold
  - Unusual internationally; property left to a spouse rolls over tax-deferred

Should Capital Gains Be Taxed Lightly?

- Optimal tax theory supports a preference no more than H-S does
- Arguments for a preference, and the doubts about each
  - Gains are windfalls, or reward waiting: but work involves sacrifice too
  - It spurs investment and risk-taking: not clearly, and there are cheaper ways to do it (Chapter 18)
  - It offsets inflation: true, but a lower rate is a crude way to adjust for it
  - Much of it is corporate income: the overall rate depends on corporate tax too (Chapter 21)
- Verdict: measured against H-S, Canada's treatment of capital gains is mixed

iClicker Question

Ana and Ben each own shares with a $10,000 gain. Ben sells and buys almost identical shares; Ana keeps hers for 10 more years. Both face the same tax rate now and later, and the shares keep growing. Why does Ana end up better off?

A) Only half of Ana’s gain will be taxed, but all of Ben’s
B) Her gain is never taxed if she holds the shares until she dies
C) She defers the tax, which works like an interest-free loan that keeps earning a return
D) She doesn’t: in the end both pay tax on the same gains
E) Her gain becomes exempt once she has held the shares long enough

Employer Benefits, Gifts and Inheritances

- Employer contributions to registered pension plans are not taxed until the pension is paid
  - Nor is the investment income the fund earns along the way
  - H-S counts both as income when they accrue, but taxing them is seen as politically infeasible
  - Employer-paid health and dental plans are also left out
- Gifts and inheritances are not taxed as income
  - Most industrialized countries tax them separately, through estate or gift taxes (the US among them)
  - Canada and Australia are the exceptions (Chapter 20)

Tax Deductions and Tax Credits

Deductions and Credits

Tax Deduction

An amount subtracted from income to arrive at taxable income

Tax Credit

An amount subtracted from the tax owed to arrive at tax payable

- Why have them?
  - Tax only net income: subtract the costs of earning it
  - Recognize unavoidable costs, such as subsistence and large medical bills
  - Encourage particular uses of income: saving for retirement, education, giving to charity
- Deciding which costs are truly unavoidable is always controversial

Deductions Change Prices

- Suppose spending on good Z is deductible and your marginal tax rate is t
  - Each \$1 spent on Z cuts taxable income by \$1 and saves t in tax
- Example: Z costs \$10 and t = 40%: each unit saves \$4 in tax, so it really costs \$6

Effective Price with a Deduction

A deduction lowers the price of Z from P_Z to $(1 - t)P_Z$

- Two consequences
  - The relative price changes, so the quantity of Z bought changes too
  - The higher your marginal rate, the more a deduction is worth to you

Main Deductions

- Child care: up to $8,000 per child under 7, $5,000 aged 7–16, $11,000 with a disability
  - Claimed by the lower-earning parent, up to two-thirds of their earned income
- Union and professional dues, and employment expenses the employer doesn’t repay
- Moving expenses when the move is to earn income
- Carrying charges: interest on money borrowed to invest
- Spousal support: deducted by the payer, taxed to the recipient
  - The payer is treated as passing the recipient’s income along
  - Child support under orders made after April 1997 is neither deducted nor taxed

Interest Deductibility and Tax Arbitrage

- Interest on consumer loans and on the mortgage on your own home is not deductible
  - Consumer interest is just the price of buying something sooner
  - Imputed rent isn't taxed, so the interest that finances the house isn't deducted
- Interest on money borrowed to earn taxable income is deductible

> **Tax Arbitrage**
>
> Borrowing with deductible interest to invest in an asset whose return is untaxed or tax-preferred

- Example: t = 40%, borrow at 10% (6% after tax), invest tax-free at 7.5%
- Mostly helps high-income people, and is hard to police

Tax-Assisted Saving Plans

Plan	Contributions	Investment income	Withdrawals
RPP, RRSP	Deductible (RRSP up to $33,810)	Untaxed in the plan	Taxed
TFSA	Not deductible ($7,000 a year)	Untaxed	Untaxed
RESP	Not deductible	Untaxed in the plan	Earnings and grants taxed to the student
FHSA	Deductible ($8,000 a year, $40,000 total)	Untaxed	Untaxed for a first home

- Home Buyers' Plan: borrow up to $60,000 from your RRSP for a first home, repaid over 15 years
- All are departures from H-S; whether they raise total saving is unclear

Tax Credits Change Prices Too

Refundable vs Nonrefundable Credits

A refundable credit is paid out even if it exceeds the tax owed; a nonrefundable credit can only reduce tax to zero

- A credit at rate x lowers the price of Z from P_Z to $(1 - x)P_Z$, the same for everyone
- A deduction's value depends on your tax rate; a credit's does not
  - A $100 credit is worth $100 to anyone who owes tax
  - A $200 deduction is worth $100 at a 50% rate, $60 at 30%
- Most federal credits are set at the lowest rate: 14% in 2026
  - So a credit is worth the same as a deduction only for someone in the lowest bracket

Main Tax Credits

Credit	Type	Why
Basic personal amount ($16,452)	Nonrefundable	Income needed to live on
Spouse, dependants, age, disability	Nonrefundable	Extra costs, less ability to pay
Base CPP and EI contributions	Nonrefundable	Required; could be deductions
Medical costs above a threshold	Nonrefundable	Large, unavoidable costs
Tuition; charitable donations	Nonrefundable	Encourage education and giving
Canada Workers Benefit (CWB)	Refundable	Top up low pay; encourage work
Canada Groceries and Essentials Benefit	Refundable	Offset sales taxes
Canada Child Benefit (CCB)	Refundable	Help with the cost of children
Boutique credits (home accessibility)	Nonrefundable	Favoured behaviour; adds complexity

Charitable Giving: Credit vs Deduction

Canada: Credit (Ontario, 2026)

For gifts above $200 in a year:

- Federal credit of 29% (33% on income in the top bracket) plus Ontario's 11.16%
- Each $1 given costs about $0.60 ($0.56 in the top bracket): much the same for every donor

United States: Deduction (Stylized)

A gift is deducted from taxable income:

- The price of $1 given is $1 - t$
- $0.76 in the 24% bracket, $0.63 in the 37% bracket
- Richer donors pay a lower price

- Canadian estimates put the price elasticity of giving above 1
  - Giving rises by more than the price falls, so charities gain more than the government loses

Credits or Deductions?

- Before 1988, Canada used exemptions (deductions) for the taxpayer, spouse, dependants and age
  - A $5,000 exemption saved $2,500 at a 50% rate but $1,000 at 20%
- The 1987 reform turned them into credits, worth the same dollar amount to everyone
- But the choice is not that simple
  - Deductions favour high earners, but higher top rates can collect the same total tax from them as a credit
  - For costs that reduce ability to pay, like unavoidable medical bills, a deduction taxes people on what they have left
  - To change behaviour, put the subsidy where people respond: a deduction gives high earners the biggest price cut, a credit gives everyone the same one

iClicker Question

Counting federal tax only: Ottawa replaces a deduction for some expense with a 14% nonrefundable credit on the same amount. Compared with the deduction, who loses?

A) Only people who owe no tax
B) People in the lowest tax bracket
C) No one, because a credit is worth the same to everyone
D) Everyone, because credits come off tax rather than income
E) People whose marginal tax rate is above 14%

Impact on the Tax Base

2024 tax year, all individual returns	$ billions
Total income	2,024
Net income, after $166B of deductions	1,856
Taxable income, after $65B more	1,791
Federal nonrefundable credits	117
Net federal tax	223
Net provincial and territorial tax, excluding Quebec	101

- Deductions take about 11.5% of total income out of the base
- Federal nonrefundable credits are worth about half the net federal tax collected

Source: CRA, Individual Income Tax Return Statistics, 2024 tax year, Table 2

Tax Expenditures

Tax Expenditure

Revenue the government gives up through preferential tax treatment: exemptions, deductions, credits, deferrals and low rates

- Exempting Z costs the same as handing $1 billion to the people who buy Z
  - One shows up on the revenue side of the budget, the other on the spending side
- Finance Canada has published estimates since 1979 (2026 report)
  - Largest personal items, 2026: RPPs $25.4B, RRSPs $17.4B, partial inclusion of capital gains $16.8B, principal residence $8.5B
- The items can't simply be added up: they interact, and people respond

Tax Expenditures: For and Against

Criticisms

- Estimates assume people wouldn't change behaviour if a break were removed
- They need a benchmark tax base, and there is no agreed one
  - One person's loophole is another's fair adjustment
- They imply that untaxed income belongs to the government

Defences

- Tax breaks are subsidies, so review them regularly like spending
- They let us compare spending through the tax system with direct spending on the same goal
- They allow benefit-cost analysis of tax breaks

The Structure of Personal Tax Rates

Rates and Brackets

- Canada has a progressive rate structure: higher incomes pay a larger share in tax
- It uses brackets: ranges of taxable income, each with its own rate

Marginal vs Average Tax Rates

- The marginal tax rate is the tax on an additional dollar of income: what matters for decisions to work, save and invest
- The average tax rate is total tax divided by total income

- The rate in a bracket applies only to the dollars that fall in it
- Federal rates ran from 4% to 29% in 1917 and 15% to 84% by 1949; there are 5 federal brackets today

Federal and Provincial Rates, 2026

Federal taxable income	Up to $58,523	to $117,045	to $181,440	to $258,482	Over $258,482
Rate	14%	20.5%	26%	29%	33%

Federal basic personal amount: $16,452

	Lowest	Top	Basic amount
NL	8.7%	21.8%	$13,094
PE	9.5%	20.0%	$15,000
NS	8.79%	21.0%	$11,932
NB	9.4%	19.5%	$13,664
QC	14%	25.75%	$18,952
ON	5.05%	13.16%	$12,989
MB	10.8%	17.4%	$15,780

	Lowest	Top	Basic amount
SK	10.5%	14.5%	$20,381
AB	8%	15%	$22,769
BC	5.6%	20.5%	$13,216
NT	5.9%	14.05%	$18,198
NU	4%	11.5%	$19,659
YT	6.4%	15%	$16,452

Ontario also has a surtax: 20% of Ontario tax over $5,818, plus 36% of Ontario tax over $7,446

Top Combined Marginal Rates, 2026

	Ordinary	Dividends	Gains
NL	54.80	46.20	27.40
PE	53.00	37.92	26.50
NS	54.00	41.58	27.00
NB	52.50	32.40	26.25
QC	53.31	40.11	26.66
ON	53.53	39.34	26.77
MB	50.40	37.78	25.20

	Ordinary	Dividends	Gains
SK	47.50	29.64	23.75
AB	48.00	34.31	24.00
BC	53.50	36.54	26.75
NT	47.05	28.33	23.53
NU	44.50	33.08	22.25
YT	48.00	28.92	24.00

- Federal plus provincial, %; dividends are eligible dividends
- Only half a capital gain is taxed, so its top rate is half the ordinary rate

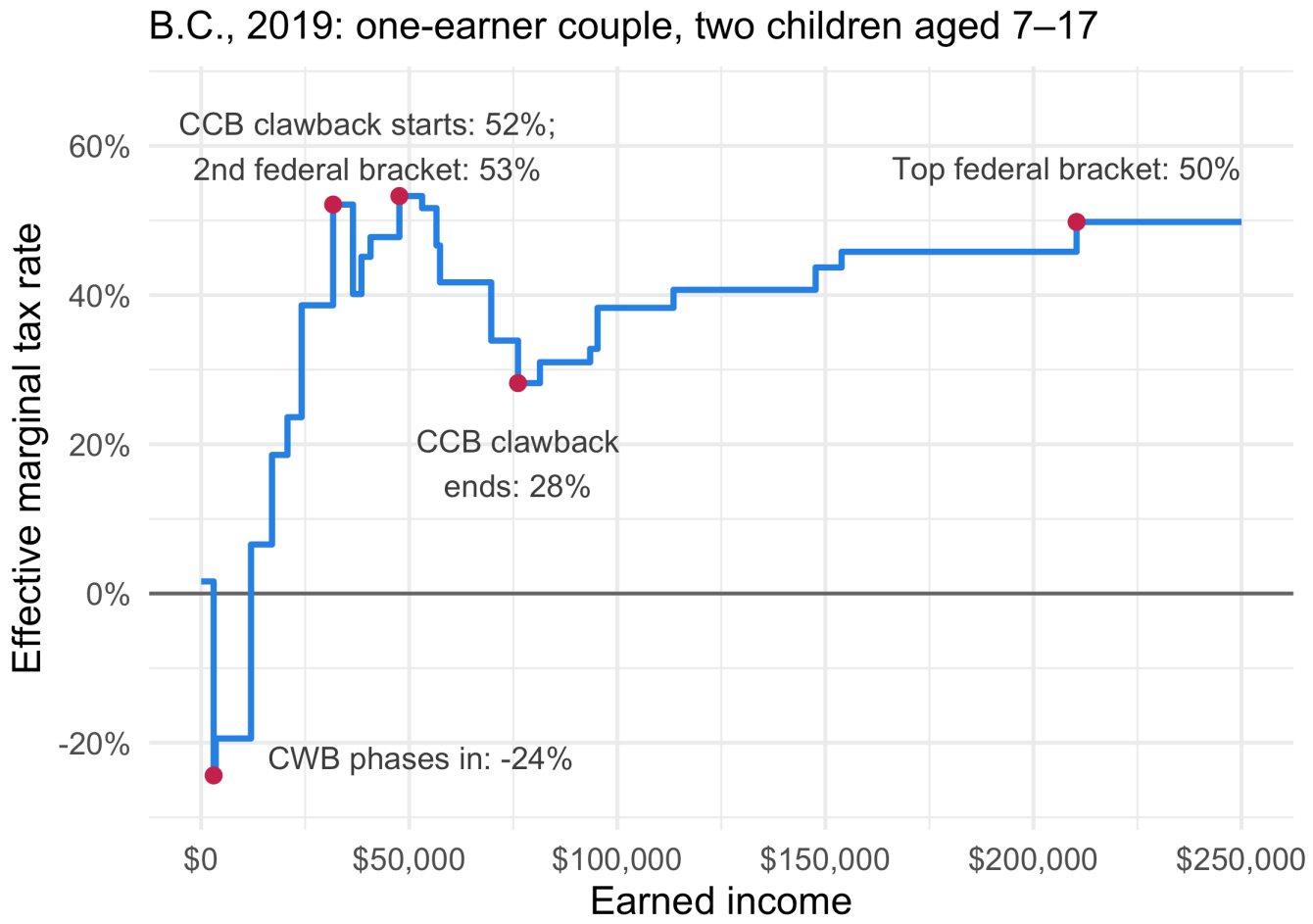
Factors Affecting Marginal Rates

- The statutory rate is the one in the law; what you keep from an extra dollar can be very different

> **Effective Marginal Tax Rate**
>
> The share of an extra dollar of earnings lost to income tax, payroll taxes and reduced benefits

- Surtaxes: Ontario is now the only province with one
- Payroll taxes: CPP 5.95% and EI 1.63% of earnings, up to annual maximums
- Clawbacks: benefits shrink as income rises, which works like a tax
  - CCB, Canada Groceries and Essentials Benefit, EI benefits, OAS
  - CWB, single person, 2026: 27% phase-in above $3,000 (up to $1,665), clawed back at 15% from $27,392 until gone at about $38,492

Marginal Tax Rates: A B.C. Example



- Negative at low incomes while the CWB phases in
- Over 50% in parts of the $30,000–$57,000 range, as the CWB, CCB and GST credit clawbacks stack on tax and payroll taxes
- The rate does not rise smoothly with income

iClicker Question

Jo is single, lives in Ontario and earns $30,000 in 2026. On an extra dollar of earnings, Jo pays 14% federal tax, 5.05% Ontario tax, 5.95% CPP and 1.63% EI, and loses 15 cents of the CWB. Counting only these, what is Jo's effective marginal tax rate?

A) 19.05%
B) 26.63%
C) 34.05%
D) 41.63%
E) 14%

Alternative Minimum Tax

- Deductions, credits and preferences can leave some high-income people paying little or no tax
  - E.g., large capital gains, dividend tax credits, tax shelters, big pension and RRSP contributions

Alternative Minimum Tax (AMT)

A second tax calculation on a broader base; you pay whichever is higher, regular tax or the AMT

- In place since 1986, and tightened in 2024. For 2026:
  - Flat rate: 20.5% on adjusted income above an exemption of $181,440 (indexed)
  - Base: all of capital gains is included, and only half of most credits count

Effective Versus Statutory Rates

- Statutory rates can mislead about who really pays, for three reasons
  - Preferences: taxable income can be much smaller than a comprehensive measure of income
  - Incidence: taxes can be shifted, so the person who pays isn't necessarily the person who bears the tax (Chapter 14)
  - Excess burden and compliance costs: taxes cost more than the revenue they raise (Chapter 15)
- So statutory rates say little about how progressive the system really is
  - Lower rates on a broader base could be just as progressive, with less excess burden and less evasion

The Flat Tax

Flat Tax

One rate applied to everyone and to every kind of income, with no deductions except a personal exemption and strictly defined business expenses

- For given revenue, the key trade-off is between the exemption and the rate
  - A bigger exemption makes average rates more progressive, but needs a higher rate
  - One rate of 25% (Alberta) to 30% (Quebec), with a $7,231 exemption, would have matched 2000 revenue (Emes and Clemens, 2001)

For

- Less excess burden and less incentive to cheat
- Simpler, so lower administration and compliance costs
- Can stay progressive through the exemption

Against

- May shift the burden from the rich to the middle class
- Hard to confirm, since it depends on who really bears the tax

- Canada moved partway: the top federal rate fell from 43% (1981) to 29% (1988) as the base broadened; a 33% top bracket was added in 2016

iClicker Question

A flat tax of 25% applies to income above a $20,000 exemption. What are the average tax rates for someone earning $40,000 and someone earning $100,000, and what kind of tax is this?

A) 25% and 25%: the tax is proportional
B) 12.5% and 20%: the tax is progressive
C) 12.5% and 20%: the tax is proportional, because there is only one marginal rate
D) 12.5% and 25%: the tax is progressive
E) 25% and 20%: the tax is regressive

Choice of Tax Unit

Three Principles for the Tax Unit

- Canada taxes the individual, and always has
  - Some countries tax the family: income is pooled and taxed on one return
- Three principles most people find appealing

Three Principles

1. The income tax should have increasing marginal tax rates
2. Families with equal incomes should pay equal taxes
3. Two people's taxes should not change when they marry or move in together: the tax should be marriage neutral

- With increasing marginal rates, no system can satisfy both 2 and 3

The Marriage Penalty

- Two married couples: Lalan and Rahi, and Emile and Florian
- Tax schedule: 10% on the first $6,000 of income, 50% above that

	Income	Tax filing alone	Couple's tax, separate	Couple's income	Couple's tax, joint
Lalan	1,000	100	12,200	30,000	12,600
Rahi	29,000	12,100			
Emile	15,000	5,100	10,200	30,000	12,600
Florian	15,000	5,100			

- Filing separately, two couples with the same income pay different tax: principle 2 fails
- Filing jointly, they pay the same, but both pay more than as singles: a marriage penalty of $400 and $2,400, so principle 3 fails

Family or Individual?

The case for the family

- The Carter Commission (1966): the family is the basic economic unit
- Families with equal incomes have equal ability to pay, so should pay equal tax
- Taxing individuals invites income splitting: shifting income to a lower-taxed spouse or child
  - E.g., "sprinkling" income from a private corporation (CCPC) to family members, limited by the tax on split income (TOSI) since 2018

The case for the individual

- Pooling taxes a second earner's first dollar at the family's top rate
  - Married women's labour supply is more elastic, so this is inefficient (Chapter 18)
- One-earner couples also get untaxed household production
- Who counts as a family? Families don't necessarily share income
- Only the individual unit is marriage neutral

iClicker Question

The tax schedule is 0% on the first $10,000 of income and 30% above that, the same for single and joint returns. Couple 1 earns $40,000 and $0; couple 2 earns $20,000 and $20,000. If each couple must file jointly once married, what marriage penalty does each pay?

A) Couple 1: $3,000; couple 2: $0
B) Couple 1: $0; couple 2: $0
C) Couple 1: $9,000; couple 2: $6,000
D) Couple 1: $3,000; couple 2: $3,000
E) Couple 1: $0; couple 2: $3,000

Taxes and Inflation

Inflation and an Unindexed Tax

Nominal vs Real Income

Nominal income is the number of dollars received; real income is what those dollars can buy

- Tax law sets many dollar amounts: bracket limits, credit amounts, clawback thresholds
- If they are fixed, inflation raises real tax burdens with no vote in Parliament
  - This happens even when inflation is fully anticipated

Indexation

Raising these amounts automatically with inflation, e.g., $10,000 becomes $10,200 when prices rise 2%

- Canada indexes to the previous year's CPI

Bracket Creep and Fixed Amounts

Bracket Creep

When incomes rise only with inflation, an unindexed progressive tax pushes people into higher brackets even though their real income hasn't changed

- Example: 10% on the first $6,000 of income, 50% above
  - Earn $6,000: tax is $600
  - Prices and income both rise 10%, to $6,600: tax is $600 + 0.5 × $600 = $900
  - Same real income, but tax rose from $600 to $900
- Fixed credit and deduction amounts also lose real value
  - Child care limits are not indexed; the basic personal amount, age amount and medical threshold are

Inflation and Capital Income

- Even a proportional tax distorts capital income when there is inflation
  - Capital gains: an asset rises 5% while prices rise 5%; the real gain is zero, but tax is due on the 5%
  - Interest: tax applies to the whole nominal rate, including the part that just offsets inflation

Real After-Tax Return (Eq. 17.1)

$r = (1 - t)i - \pi$, where i is the nominal interest rate, t the tax rate and π inflation

- With i = 6%, π = 3% and t = 50%: $r = 0.5 \times 6\% - 3\% = 0$
  - If inflation rises to 6% and i to 9%: $r = 4.5\% - 6\% = -1.5\%$
- Borrowers deduct nominal interest, so inflation lowers their tax

Inflation and Interest: An Example

- You have \$100 and a bike costs \$100. Lend the \$100 for a year, pay 50% tax on the interest, then buy the bike

	$i = 6\%, \pi = 3\%$	$i = 9\%, \pi = 6\%$
Interest earned	\$6.00	\$9.00
Tax at 50%	\$3.00	\$4.50
Money after tax	\$103.00	\$104.50
Bike price next year	\$103.00	\$106.00
Bikes you can buy	1.000	0.986
Real after-tax return	**0%**	**−1.4%**

- Only about \$3 of the interest is a real gain; the rest just keeps up with the bike's price
  - The tax takes 100%, then 150%, of the real gain

iClicker Question

You lend money at a nominal interest rate of 8%. Inflation is 4% and your tax rate on interest income is 25%. What is your real after-tax rate of return?

A) 2%
B) 3%
C) 4%
D) 6%
E) It depends on whether the 4% inflation was anticipated

Indexation in Canada

Period	Treatment
1949–1973	Not indexed: the CPI rose 72% from 1949 to 1971, and income tax went from 16% to 30% of government revenue
1974–1983	Full indexing of brackets and exemptions
1984–1985	Indexing capped at 6% and 5%
1986–1999	Only inflation above 3% a year; nothing after 1992, when inflation fell below 3%
2000 on	Full indexing of brackets and credits restored

- Capital income has never been indexed: measuring real gains is too complex
- Some amounts are still fixed, e.g., child care limits and several credits (Hansen, Mescall and Purse, 2023)

Should Taxes Be Indexed?

Against

- Without indexing, governments must revisit the tax law regularly, a chance to fix other problems
- Indexing slows revenue growth, which may push governments toward less equitable taxes

For

- A stable, predictable tax law, with fewer chances for mischief
- No tax increases without a vote: real rates shouldn't rise by stealth
- The stakes are large: by 1984, indexing had cut federal and provincial revenue by $19.5B, or 40%
- Without indexing, low-income families lose most: fixed allowances and clawback thresholds hit them first

Treatment of International Income

Residence-Based Taxation

- Canada taxes on residence: people resident here 183 days or more are taxed on their worldwide income
- To avoid double taxation, there is a credit for foreign taxes paid, up to the Canadian tax on that income
  - E.g., Canadian tax on income earned in Germany is $7,000 and German tax was $5,500: pay $1,500 to the CRA
- Canadians who live abroad owe no Canadian tax on their foreign income
- The US taxes its citizens on worldwide income wherever they live, with a similar credit

Territorial vs Global Systems

A territorial system taxes citizens living abroad only in the host country; a global system taxes citizens on all their income wherever they live

Territorial vs Global Systems

Equity

- A Canadian and an American live and work in Hong Kong with the same income
  - The Canadian pays only Hong Kong tax; the American may also owe US tax
- Canada treats Hong Kong residents alike, but Canadian citizens differently depending on where they live
- Should horizontal equity be defined by residence or citizenship? No system satisfies both

Efficiency

- A global system can distort production: US firms abroad pay more for American staff, who owe US tax
- A territorial system can distort location: Canadians can cut their tax by moving abroad
- Which distortion costs more is unknown; the territorial system is easier to enforce

Summary

Summary

- The personal income tax is the largest single source of tax revenue in Canada
- Tax owed runs from total income, through deductions, to taxable income, then rates and credits
- Haig-Simons income is the benchmark; Canada departs from it for capital gains, pensions, gifts and imputed rent
- Deductions are worth more at higher rates, credits the same to all; both change prices
- Effective marginal rates differ from statutory ones because of payroll taxes and clawbacks
- No tax unit gives rising rates, equal tax for equal-income families and marriage neutrality; Canada taxes the individual
- Without indexation, inflation raises real tax burdens; capital income is still taxed on nominal returns
- Canada taxes residents on worldwide income; the US taxes citizens

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